

Surya Teja Mothukuri

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Summary

Data Scientist & ML Engineer building end-to-end ML and GenAI systems across healthcare and operations. Architects RAG pipelines, agentic AI workflows, and LoRA fine-tuned LLMs (LLaMA, GPT-4) plus classical ML — XGB/LightGBM risk models on 90k+ encounters, demand forecasting on 10M+ transactions cutting spend by 18%, saving ~\$1.7M annually.

Technical Skills

Programming & Scripting: Python, SQL, R, C, VBA(Excel Macros), MATLAB, DAX, Bash, PowerShell
AI and NLP: LangChain, RAG, Agentic AI, LoRA, GPT-4, Vector DBs, FAISS, LLaMA, Hugging Face, Transformers, RLHF
Machine Learning & Modeling: MLflow, PySpark, CNNs, GANs, VAEs, TensorFlow/Keras, PyTorch, XGBoost, LightGBM, Scikit-learn, Optuna, ARIMA, ONNX, Anomaly Detection, Feature Engineering, SHAP, OpenCV, Librosa, OpenSMILE
Data Handling & Analysis: ETL Pipelines, dbt, Airflow, Databricks, Pandas, NumPy, SciPy, Polars, Statistical Modeling.
Databases & Tools: AWS (VPC, EC2, Auto Scaling, SageMaker), MySQL, MongoDB, SQLite, Tableau, Power BI, Git, Docker, Advanced Excel.

Professional Experience

Data Scientist - Operations Oct 2024 – May 2026
Indiana University Bloomington

- Reduced forecast error (MAPE) by 80% and cut excess procurement spend by 18%, saving ~\$1.7M annually, by engineering an XGBoost + ARIMA ensemble for SKU-level demand forecasting across 10M+ transactional records.
- Cut refresh latency by 40% with dbt + SQL ETL pipelines unifying sales, event, and inventory data for real-time BI analytics.
- Built 3 Power BI dashboards with inventory turnover and anomaly detection, driving 25% faster decisions for 10+ stakeholders.
- Engineered Python + SQL alerting for low-stock and abnormal demand, cutting waste by 20% and holding costs by 15%.

Data Scientist Intern May 2025 – Aug 2025
Indiana University School of Medicine

- Unified 10+ raw EHR exports across 90k+ encounters into a 200+ feature ICD-labeled delirium dataset via Python + SQL.
- Tracked RF and XGBoost runs in MLflow with CV and held-out testing, achieving AUROC 0.82 (Brier 0.14, slope 0.98).
- Boosted high-risk recall by 23% at fixed precision via Optuna-tuned hyperparameters and calibration-aware thresholding.
- Cut model refinement time by 31% by surfacing key delirium risk drivers via SHAP analysis for targeted feature iteration.

Automation Data Engineer Oct 2023 – July 2024
TATA Consultancy Services Ltd.

- Saved ~180 staff hrs/month by deploying 15 Python + PowerShell workflows via Windows Task Scheduler with automated failure alerting and reliability reporting to client management.
- Cut refresh times to under 30 min and compute costs by 33% by optimizing Snowflake ETL with dbt models for daily ingests.
- Reduced reporting rework by 25% by implementing schema validation and exception routing via a GitLab CI/CD pipeline.
- Mitigated 55% of pipeline failures by implementing structured logging and exception handling across 15 automated workflows.

Computer Vision Intern Jun 2022 – Dec 2022
FORD Motor Company

- Rebalanced 90:10 to 60:40 on 300k images via CycleGAN + OpenCV synthetic defect augmentation in PyTorch + TensorFlow.
- Cut false negatives by 50%, hitting 0.91 recall via defect taxonomy alignment and ResNet-50 fine-tuning on augmented data.
- Surfaced recurring failure modes via misclassification error analysis, enabling targeted augmentation and retraining strategies.

Additional Experience

Graduate Associate Instructor - CSCI P 556 Applied Machine Learning Aug 2025 – May 2026
Luddy School of Informatics, Computing and Engineering, Indiana University Bloomington

- Mentored 200+ students in lab sessions and office hours on ML pipelines, model evaluation, and core ML and DL concepts.

Projects

Autonomous Multi-Agent Bargain Spotter using RAG-Enhanced LLMs

- Improved deal relevance by 40% by orchestrating GPT-4 multi-agent scanning via LangChain with FAISS-grounded retrieval.
- Automated 95% of offer notifications by developing a real-time RAG + GPT-4 pipeline, eliminating manual deal monitoring.

Generative AI in Online Discourse: Reddit Social Media Analysis

- Analyzed 100k+ Reddit posts with VADER, BERTopic, and custom embeddings to track sentiment and topic shifts on GenAI.
- Improved topic coherence and reduced cluster noise by 22% via an NLTK-based normalization pipeline for noisy GenAI text.

Education

Indiana University Bloomington Aug 2024 – May 2026
Master of Science in Data Science, CGPA : 3.9/4.0

Indian Institute of Information Technology Aug 2018 – May 2023
Integrated Masters and Bachelors in Electronics and Communication Engineering, CGPA : 3.7/4.0